

OTFS-Based Integrated Sensing and Communication System for Vehicular Networks

Calin Cristian Dinga Pastae

Bachelor's Thesis – Universidad Carlos III de Madrid (2024)

Abstract

This document provides an English technical summary of the Bachelor's Thesis "*Estimación de canal conjunta para escenarios vehiculares con ISAC*". The work proposes an uplink Integrated Sensing and Communication (ISAC) system for vehicular networks using Orthogonal Time Frequency Space (OTFS) modulation, enabling joint communication and sensing under high-mobility conditions.

Motivation and Context

Future vehicular networks must support reliable data transmission while simultaneously enabling sensing functionalities such as vehicle localization and velocity estimation. High mobility and Doppler effects severely degrade the performance of conventional OFDM-based systems.

Integrated Sensing and Communication (ISAC) addresses these challenges by unifying communication and sensing within a single waveform. OTFS modulation, operating in the delay–Doppler domain, provides robustness against fast channel variations and is therefore well suited for vehicular scenarios.

System Model

The considered scenario consists of a Road Side Unit (RSU) and multiple vehicles transmitting uplink signals within the RSU coverage area. After an initial signaling phase, vehicles periodically transmit OTFS-modulated ISAC signals.

From the received signal, the RSU aims to recover the transmitted message, estimate the channel, and infer sensing-related parameters such as delay and Doppler, which can be mapped to distance and velocity.

Figure 1 illustrates the vehicular communication and signaling scenario.

OTFS for Integrated Communication and Sensing

OTFS maps information symbols directly into the delay–Doppler domain, where wireless channels exhibit a sparse and more stable structure. Each propagation path is characterized by a specific delay and Doppler shift, which can be directly associated with sensing parameters.

This representation enables joint communication and sensing without additional radar waveforms or signaling overhead, provided that suitable pilot strategies are employed.

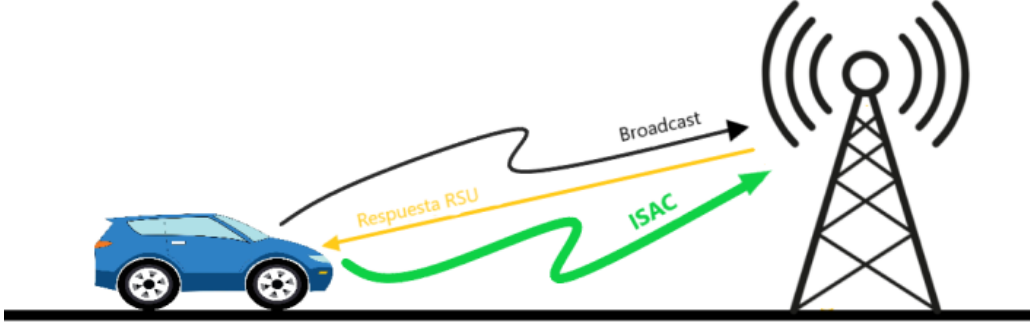


Figure 1: Vehicular V2I scenario with uplink ISAC transmission.

Pilot Design and Multi-User Identification

The main contribution of this work is a structured pilot allocation scheme in the delay–Doppler domain that enables multi-user identification and joint channel estimation.

The global delay–Doppler grid is partitioned into disjoint blocks, each assigned to a specific vehicle. A high-power pilot symbol is placed within each block. After propagation through the channel, the pilot displacement in the received grid allows the RSU to identify the transmitting vehicle and estimate the associated delay and Doppler parameters.

The block-based grid structure and pilot placement strategy are shown in Figure 2.

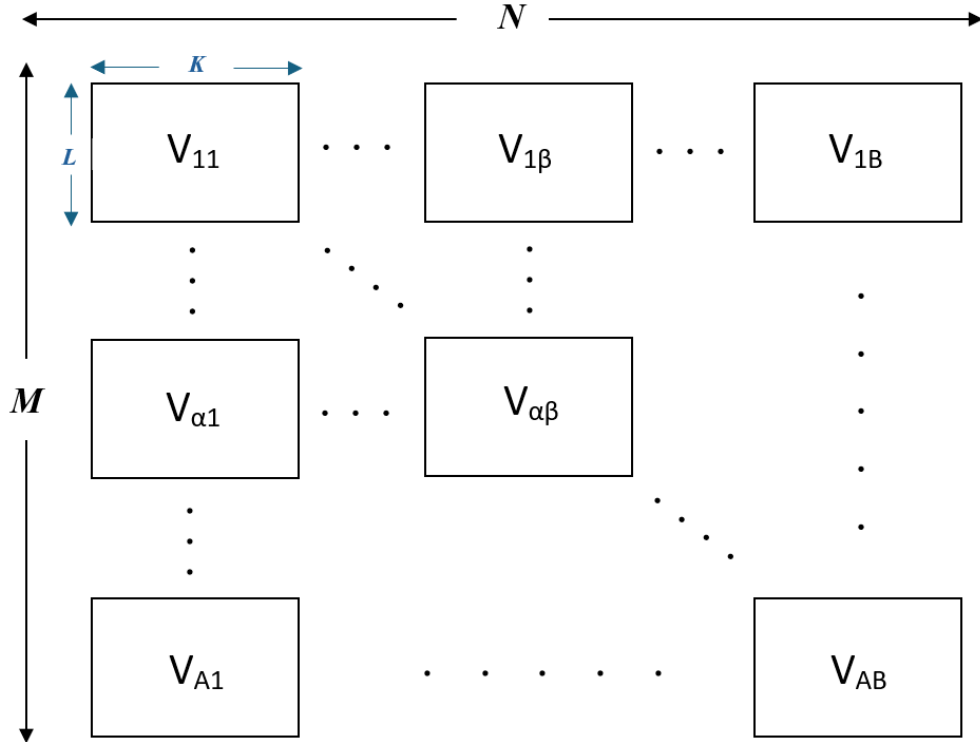


Figure 2: Block-based delay–Doppler grid allocation and pilot placement.

Receiver Processing and Parameter Estimation

At the receiver, the RSU processes the received OTFS signal in the delay–Doppler domain to jointly estimate the channel and sensing parameters. The displacement of the pilot symbol within the assigned block is exploited to identify the transmitting vehicle and extract the corresponding delay and Doppler indices.

These indices are directly mapped to physical quantities such as propagation delay and relative velocity, enabling vehicle localization and motion estimation. Once the channel parameters are estimated, the received symbols are equalized to recover the transmitted data message.

Figure 3 illustrates the receiver-side processing pipeline used for joint communication and sensing.

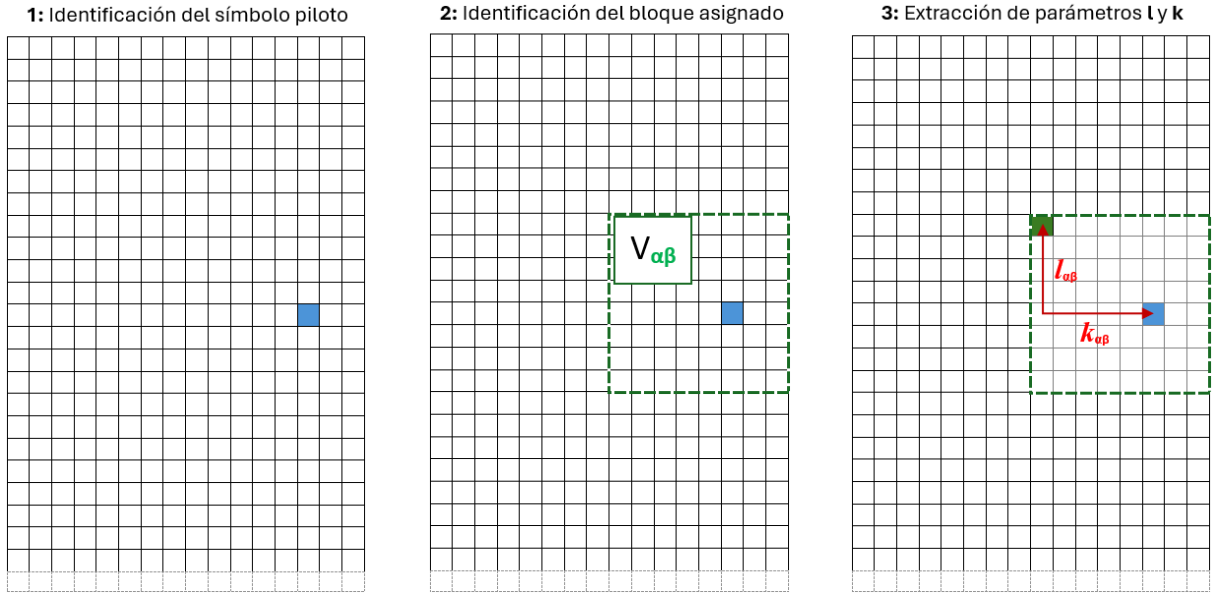


Figure 3: Parameter estimation process from the transmission matrix received in the delay-Doppler domain of a given vehicle $\alpha\beta$.

Performance Evaluation

The proposed system is evaluated through MATLAB-based simulations under AWGN and varying SNR conditions. Performance metrics include Bit Error Rate (BER) and robustness of pilot detection.

Results show reliable performance at medium-to-high SNR values. At low SNR, the system exhibits strong dependence on pilot power, revealing a trade-off between pilot energy, spectral efficiency, and sensing reliability.

Figure 4 illustrates BER performance for different pilot power levels.

Conclusions and Future Work

This work demonstrates that OTFS modulation combined with structured pilot placement enables effective integrated communication and sensing in vehicular networks. The proposed approach supports multi-user operation while maintaining low receiver complexity.

Future work may include adaptive pilot strategies, machine-learning-assisted channel estimation, and validation using real hardware platforms such as SDR testbeds.

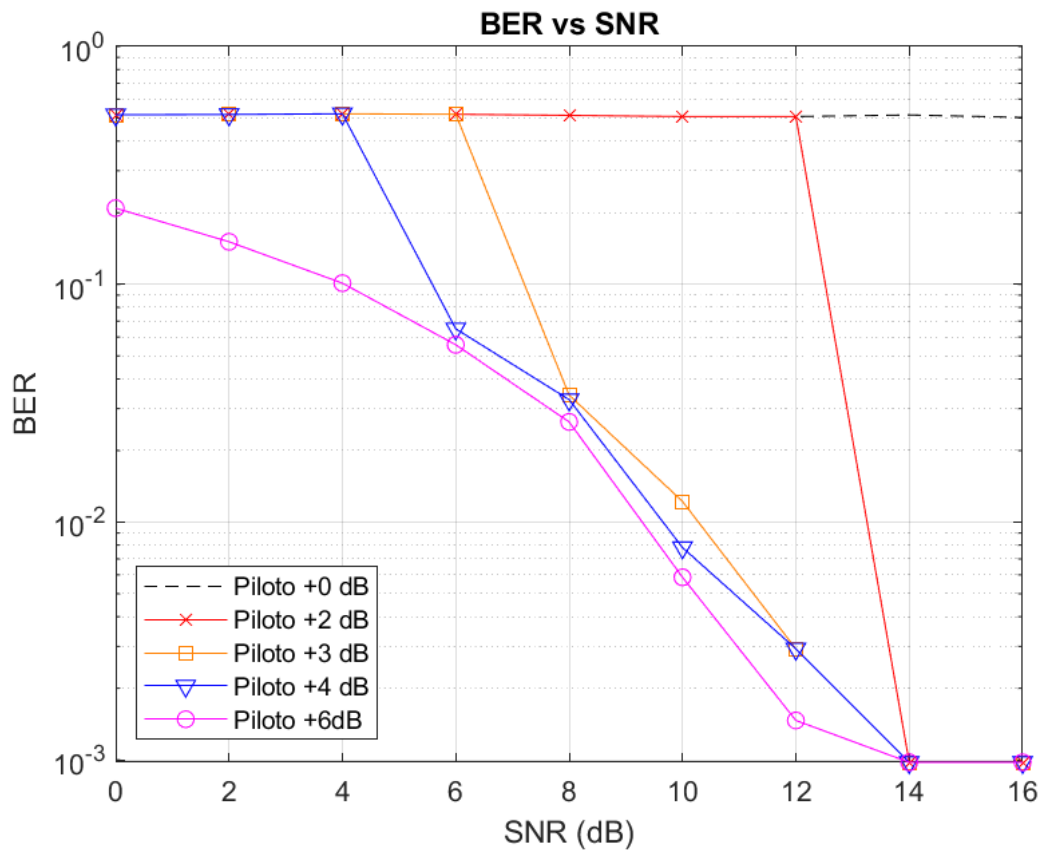


Figure 4: BER versus SNR for different pilot power allocations.